Investigating the layer-wise interpretability of the cross-modal SSL model

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Abstract

Recent advancements in the field of ASR are often met with rising curiosity about its speech-processing mechanism. While the E2E approach guarantees data-efficient, accurate judgment, the linguistic mechanics behind the identification process are not so clearly laid out. Since this ambiguity limits its applicability to related fields and further advancement, a growing body of work has focused on identifying the phonetic content embedded in its processing unit. Research in this endeavor has hitherto successfully established so-called ASR model probing methods. Meanwhile, another recent trend in speech processing technology is incorporating cross-modal training objectives to leverage the benefits of different modalities. Although several works ([3],[4]) insinuate the linguistic capabilities of these multi-modal models, exploring the phonetic content behind their hidden representation is an untapped area of research. Under this context, this paper aims to expand the application of the established probing methodology to multi-modal models.

Introduction

Layer-wise probing is a research topic often applied to self-supervised E2E models. SSL models like Wav2Vec2.0 acquire useful representations of the data during training that are later utilized to pin down characteristics of the unknown test data. Since ground-truth target labels are unprovided, this learning procedure is of a completely autonomous, model-internal nature. The lack of revealment prior to output generation amounts to a black box problem. To overcome this issue, probing sheds light on the properties of intermediate output generated at each layer. By training a classifier with a hidden output-ground truth label pair, the generalizability of the classifier to unseen data can calculate the layer's contribution to the final decision-making process. [1] and [2] use MLP probing to locate the phonetic content of each of the 12 transformer layers in wav2vec2.0.

While Wa2Vec2.0 is trained on a singular objective to recognize the transcript of speech, VG-HuBERT of [3] and [4] is trained to match the relevant image and audio data. VG-HuBERT assumes a simple dual-encoder transformer architecture whose audio encoder is identical to the one in wav2vec2.0. Training of both audio and image encoder is carried out in a completely self-supervised fashion. Hence, there is no part of the model-building process that directly relates to linguistic interpretability. Nevertheless, it exhibits word-segmentation as well as syllable-segmentation capabilities that vary across different layers.

Such language-related characteristics prompt a further need to examine the phonetic content of each layer beyond simple segmentation. In light of this understanding, the current work will use the methodology of [1] and [2] to cross-modal SSL model in [3] and [4]. The goal of this endeavor is two-fold. Our first aim is to find whether the phoneme discriminability of each layer is consistent across singular and cross-modal. On top of this model-wise comparison, we would like to limit the focal point of our analysis

to where word-level and syllable-level performances diverge.

Method

As noted earlier, the basic implementation guideline follows [1]. The work employs MLP classifiers trained to predict phoneme labels from phoneme-averaged embeddings. Accordingly, we have trained 5 classifiers in total, each accounting from layer 6 to 10. The reason for this selective approach will be expounded below.

Dataset for both training and testing is from the English read-speech corpus TIMIT. This dataset is particularly useful for phoneme-wise analysis, as it entails detailed timing information of each spoken phoneme. From the phoneme-utterance ratio inferred from 'phonetic detail', one can group the time frames relevant for each phoneme. Since the number of time frames varies for each instantiation, we took the average of all pertinent embedding values. In this way, the classifier trained on the averaged embedding of associated time frames can be used to predict phoneme labels with an input obtained in the same manner.

The **configuration of multi-layer perceptron (MLP)** follows the default configuration provided by the scikit-learn, except for the 200 neurons hidden unit size. It uses a ReLu activation for the hidden layers, while a logistic activation function is used for the final output layer. Each MLP classifier is trained with hidden output from a designated layer. When trained to examine the phonetic content of the 6th layer, for example, an embedding extracted from the 6th layer of the audio encoder is used with a paired phoneme label.

Extraction of intermediate output was viable referring to a code[1] provided by the author of [3] and [4]. The cross-modal model itself was also downloadable for usage. The MLP training pipeline is schematized in graph 3.

The phoneme set used for the classifier follows the transcription unit of TIMIT except for the start and the end token *#h*. The reason for this exclusion comes from the probabilistic nature of MLP classification. As the classifier outputs a token with the highest probability, *#h* present in every utterance will have the highest exposure rate and condition the classifier to associate every input with it regardless of the nature of the sound segment. To avoid falling into this pitfall, we have effectively removed all sentence markers. The followings are 60 phonemes used for training.

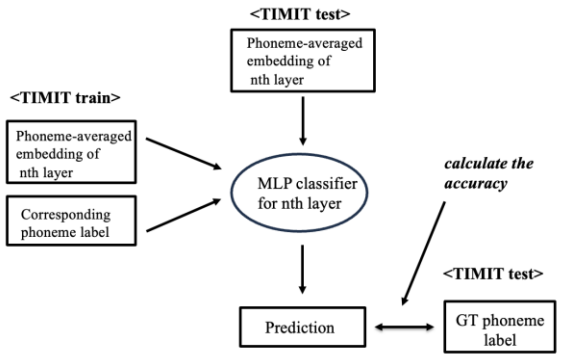
Table1

Phonemes	'aa', 'ae', 'ah', 'ao', 'aw', 'ax', 'ax-h', 'axr', 'ay', 'b', 'bcl', 'ch', 'd', 'dcl', 'dh', 'dx', 'eh', 'el', 'em', 'en', 'eng', 'epi', 'er', 'ey', 'f', 'g', 'gcl', 'hh', 'hv', 'ih', 'ix', 'iy', 'jh', 'k', 'kcl', 'l', 'm', 'n', 'ng', 'nx', 'ow', 'oy', 'p','pau', 'pcl', 'q', 'r', 's', 'sh', 't', 'tcl', 'th', 'uh', 'uw','ux', 'v', 'w', 'y', 'z', 'zh'
MOA (classified by manner of articulation)	Stops: = ['b','d','g','p','t','k','dx','q','bcl','dcl','gcl','pcl', 'tcl','kcl'] Affricates: ['jh','ch'] Fricatives: ['s','sh','z','zh','f','th','v','dh'] Nasals: ['m','n','ng','em','en','eng','nx'] Semivowels: ['l','r','w','y','hh','hv','el'] Silence: ['pau','epi'] Vowels: = ['iy','ih','eh','ey','ae','aa','aw','ay','ah','ao','oy','ow','uh','uw','ux','er','ax','ix','axr','ax-h']
POA (classified by place of articulation)	Velar: ['g','k','ng','eng','gcl','kcl'] Glottal: ['q','hv','hh'] Bilabial: ['b','p','m','em','bcl','pcl'] Post alveolar: ['jh','ch','sh','zh','r'] Labial velar = ['w'] Dental: ['th','dh'] Palatal: ['y'] Labio dental: ['f','v'] Alveolar: ['d','t','dx','nx','n','s','z','en','l','el','dcl','tcl']

[1] https://github.com/jasonppy/syllable-discovery

In Table1, phonemes are additionally grouped according to their place and manner. The classification is noteworthy, as we will later derive analysis per category

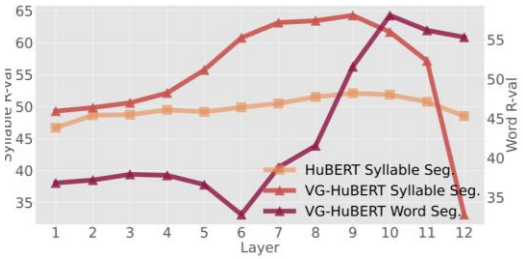
Graph3



Experiment

Locus of analysis: The previous section noted that we have limited our analysis to five middle layers. The choice of this selection relates to the performance trend observed in both VG-HuBERT and wav2vec2.0 models. For cross-modal training, this interval is where the trade-off between syllable-level and word-level detection takes place (Graph 1). For uni-modal training, phoneme detection performance reaches its peak between these layers (Graph 2) .

Graph1



Thus, examining the internal properties of these intermediate layers particularly bears importance in that it could establish the direct comparison between single-encoder and dual-encoder architectures. If the peaking trend in graph 2 remains in VG-HuBERT, the image encoder may not exert enough influence to interfere with audio

Graph2

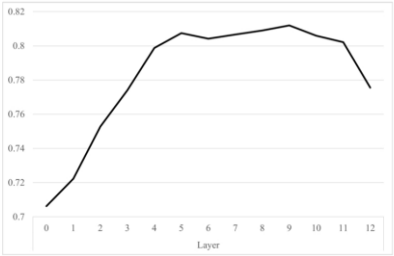


Figure 2: Average phoneme label accuracies per layer

representation learning. If the trend gets disrupted, the opposite may hold. Moreover, examining the cause behind performance disagreement of different granularities was explicitly part of a future work written out in [3]. These factors foster special attention to given five mid-to-late layers.

Level of analysis consist of the most elementary phoneme accuracy performance, MOA (manner-wise) accuracy for overall and per category, and POA (place-wise) accuracy for per category. POA accuracy was not calculated for overall scores as it was a categorization for consonants only. We have also analyzed featural distance between the reference and hypothesized phonemes. Based on PanPhon module [5], each phoneme has been identified with 24 binary phonological featural values. A single mismatch incurs 1/24 violation, whereby a lack of correspondence in insertion and deletion amounts to a distance score of 1. In order to implement this module, we have to convert the TIMIT phoneme to an ipa symbol. Below shows the mapping table.

Table2

TIMIT Phoneme: IPA	**'aa':**'ɔ','**ae':**'æ','**ah':**'ʌ','**aw':**'aʊ','**ax':**'ə','**ax-h':**'ʌ','**axr':**'ər','**ay':**'aɪ', **'b':**'ɓ','**bcl':**'ɓ','**ch':**'ʧ','**d':**'ɗ','**dcl':**'ɗ','**dh':**'ð','**dx':**'ɾ','**eh':**'ɛ','**el':**'əl', **'em':**'əm','**en':**'ən','**eng':**'ɪŋ','**epi':**'', **'er':**'ɜr','**ey':**'eɪ','**f':**'f','**g':**'g','**gcl':**'g',

	'hh':'h',**'hv':**'ɦ',**'ih':**'ɪ',**'ix':**'ɪ',**'iy':**'i',**'jh':**'ʤ' ,**'k':**'k',**'kcl':**'k',**'l':**'l',**'m':**'m', **'n':**'n',**'ng':**'ŋ',**'nx':**'n',**'ow':**'oʊ',**'oy':**'ɔɪ',**'p' :**'p',**'pau':**'',**'pcl':**'p',**'q':**'ʔ',**'r':**'ɹ', **'s':**'s',**'sh':**'ʃ',**'t':**'t',**'tcl':**'t',**'th':**'θ',**'uh':**'ʊ', **'uw':**'u',**'ux':**'u',**'v':**'v',**'w':**'w',**'y':**'j', **'z':**'z',**'zh':**'ʒ'

The accuracy for featural distance is calculated as a ratio of matched features. The utterance of the TIMIT test set uses 59105 phonemes in total, which is a featural distance score granted if entire phoneme pairs are in disagreement. Under these circumstances, accuracy = 1 – (score/59105).

Results

We first examine the **layer-wise phoneme accuracy**. To one's dismay, the accuracy is far below the expected rate, recorded at around 10% on average. There is no outstanding trend observed among layers. We attributed this result to allophonic specification of the TIMIT phonemes. The phone set in use made detailed sub-phonemic distinction, which could have worked in disfavor of aiming for a high accuracy score. To round off the decision boundary, we have also recorded accuracy based on the place and manner of articulation.

Table3

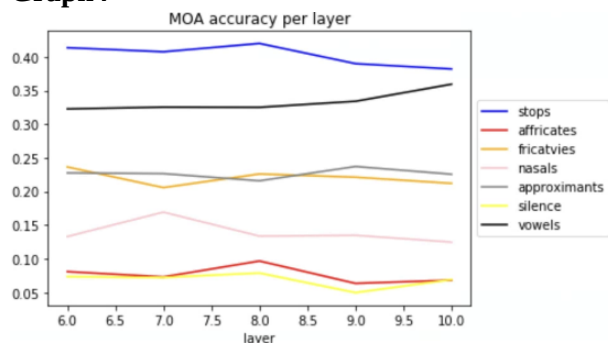
layer	Phoneme accuracy
6	10.41%
7	10.42%
8	10.35%
9	10.30%
10	10.35%

Table4

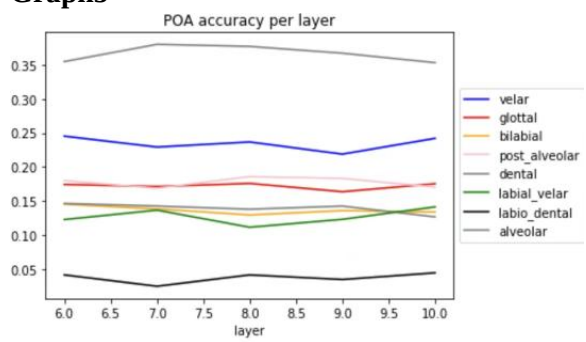
layer	(overall) MOA accuracy
6	30.37%
7	30.17%
8	30.38%
9	29.96%
10	30.29%

Although table4 shows improved results to table3, we further need to navigate the origin of improvement among different manner categories. Graph 4 and 5 each record which place and manner category has the higher or lower identifying rate within the same layer.

Graph4



Graph5



Overall, the trend held that the larger the phoneme inventory of the category is, the higher the accuracy. This stems from a rather language-agonistic statistical impact, as the larger phoneme set coverage ensures higher frequency and chances for rightful classifications. Underperformance of bilabial, outperformance of stops to vowels, and underperformance of nasals were exceptions to these rules. It was also surprising to note that silences were often incorrectly identified. This leads to a puzzling conclusion, as one would expect the model to be at least discernible to the most direct indicator of segmentation.

The drop in accuracy rate among stops, nasals, and alveolar categories resonates with findings on

loss of linguistic content in the latter layers of the transformers [2]. Even so, the trend is not consistent enough to make a concrete statement.

Discussion

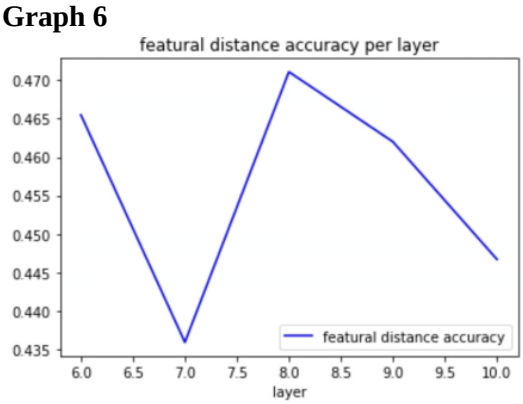
Perhaps the failure of the classifier's performance was an expected outcome from the get-go. Revisiting graph 1, the recorded accuracy for both syllable and word segmentation never surpasses the mediocre range of 0.65. Since recognizing the phonetic trait of sound is a more complex task than demarcating a syllable boundary, the phoneme identification rate being limited to under 0.11 may as well be a corollary. In the end, even if the dual-encoder architecture is versed in carrying out surface-level lingua tasks, we conclude that cross-modal training is not ultimately conducive to deriving better linguistic insights.

Before ending the paper, we would like to give an exemplary error layout among the classifiers that shows the highest featural performance and phoneme categories displaying variation outside of a pure statistical influence. Table 5 shows the feature distance classifier of each layer incurred. The score and accuracy derived based on phonological feature congruity is the most truthful reflection of how much phonological knowledge each classifier holds. The reason is that featural distance encompasses all aspects of a phoneme, whereas by-category classification taps into only a single aspect. That being said, one may infer that the layer 8 hidden output is the most phonologically truthful.

Table 5

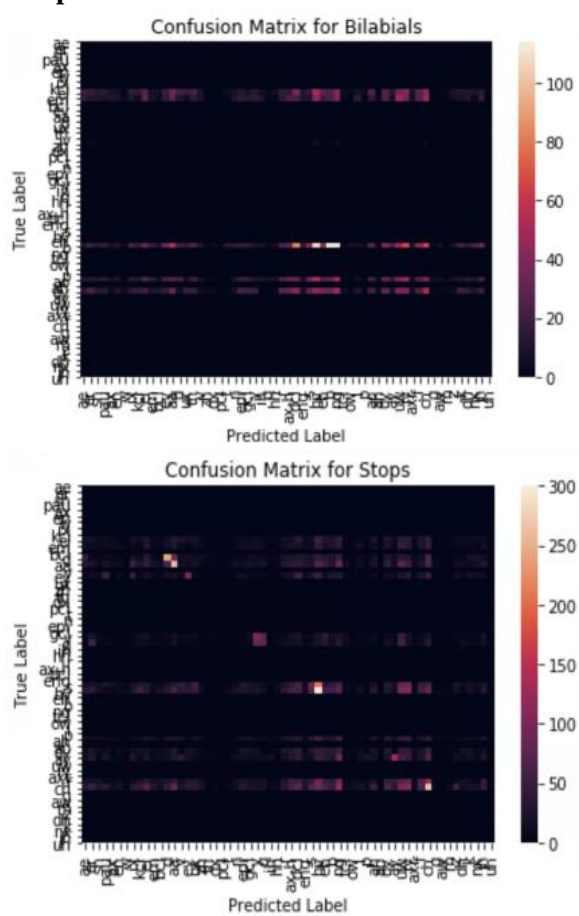
layer	Sum of featural distance among 59105 phoneme pairs
6	32487 / 46.55% accuracy
7	34288 / 43.59% accuracy
8	32145 / 47.11% accuracy
9	32697 / 46.20% accuracy
10	33629 / 44.67% accuracy

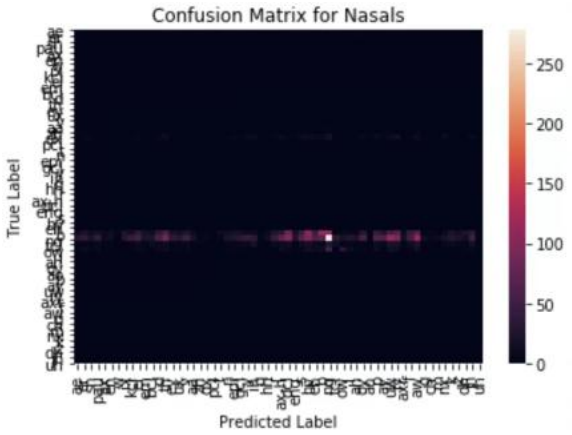
Graph 6



In the previous section, bilabial, stops, and nasals were three categories of phonemes whose performances in classifiers were not merely swayed by the statistical influence. Hence, it is worthwhile to examine how phonemes within these categories are outputted by the layer 7 MLP.

Graph7





Graph7 compares the phoneme classification results of layer 7 in more detail. A distinction to be made between different phoneme groups is that stops have a higher tendency to achieve a matching rate compared to nasals and bilabials. This shows that by-category evaluation not only assesses the rough grouping of sounds on a given phonological feature but also indicates that sounds of that nature are more prone to be accurately classified. In this context, the underperformance of bilabials is the weakest with their most spread-out prediction, while the nasals come second with their somewhat salient groupings.

In the end, we can attest to the validity of our analytical criteria and further investigate what makes under- or out- performance of a particular phonological category viable. An abrupt sound characteristic tied to stops, for instance, may make it more salient for distinction. Liquid sounds like nasal, on the other hand, will result in the opposite. This is just one hasty prediction, however, as the same argument does not hold for vowels. Pinning down the traits of sound that make it more machine-friendly shall be attended to in future work.

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